Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Robinson Medina (TNO)	**Email**	Robinson.Medina@tno.nl
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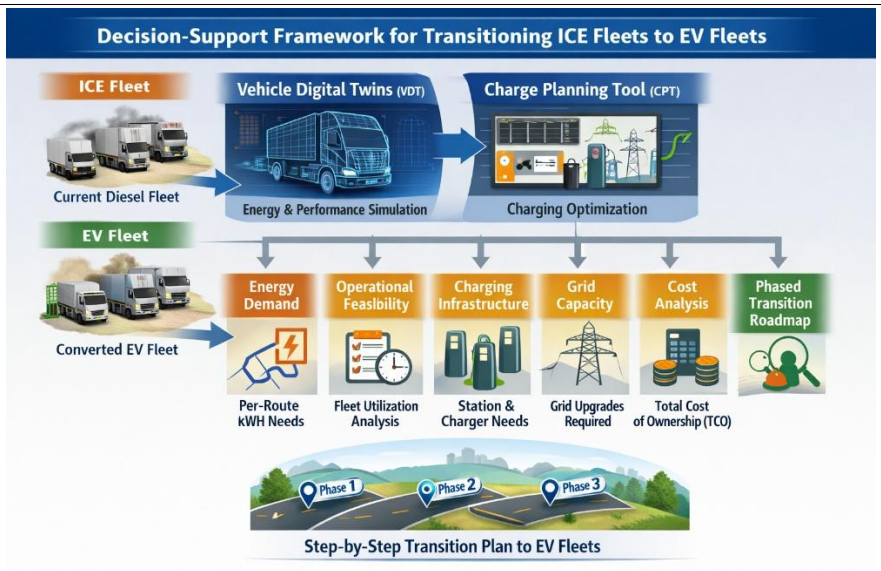
2. Technology Overview

Title of Proposed Technology	ICE to EV Fleet decision conversion framework for commercial operations			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-generation Energy (), Bio-health ( ), Aerospace ( ), Quantum ( ), Advanced Robotics & Manufacturing ( ), AI ( ), Cybersecurity (), Hydrogen ( ), Semiconductor materials ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Energy & Resources	
	Medium Category	Medium Category	Medium Category	Small Category
	ITS & Telematics	Telematics Application Services	Power IT	Intelligent power grid platform Technology
Concept of Proposed Technology	Core concept: A decision-support framework that combines Vehicle Digital Twins (VDT) and a Charge Planning Tool (CPT) to perform end-to-end assessments for converting internal combustion engine (ICE) commercial fleets to electric vehicle (EV) fleets.			

	 The framework quantifies per-route energy demand, fleet operational feasibility, charging infrastructure needs, grid capacity requirements, and total cost of ownership (TCO), and produces phased transition roadmaps. Key elements: 1) Vehicle & Battery Digital Twins that simulate drive cycles, ambient conditions, payloads, gradients and auxiliary loads to predict energy consumption, battery temperature/SoC/SoH evolution, and realistic charging power profiles under aging and uncertainty to determine energy needs of current operations. 2) Charge Planning Tool that optimizes depot charging schedules and charger mix (AC/DC power levels), while respecting depot grid limits, dynamic tariffs, and operational constraints (shift times, dwell windows, routing). 3) Validation using laboratory data (battery/vehicle test benches) and pilot fleet datasets to calibrate twins and verify planning outcomes.
Rationale for Joint R&D Project	**Need**: A company seeking advanced tools to assess the transition of its fleet from internal combustion engine (ICE) vehicles to electric vehicles (EVs). The company will contribute a real-world use case, enabling the evaluation of operational feasibility, charging requirements, and associated costs during the transition. In collaboration, TNO will co-develop and tailor digital twin-based optimization and charge planning tools to address the company's specific operational context and needs.

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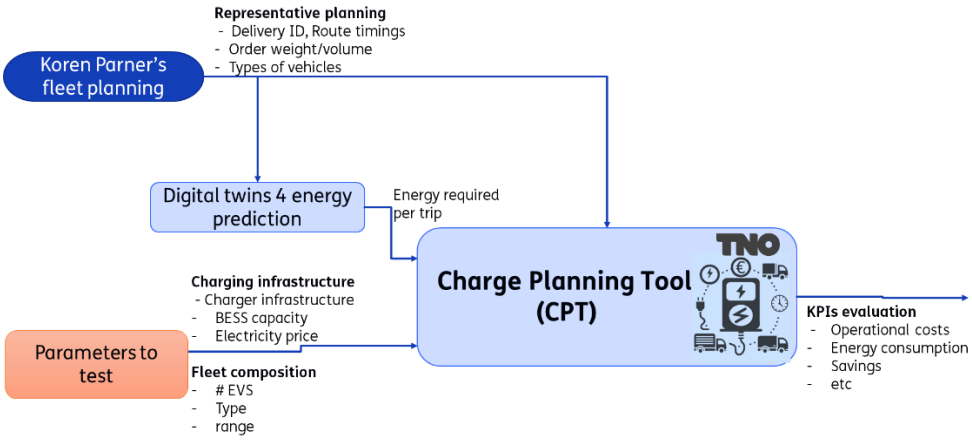

Why TNO?: TNO has a long track record on developing technologies for electrified powertrain solutions. Among others, TNO has developed several tools for the operation and estimation of usage of electric trucks in the context of commercial operations.

Collaboration rationale: The Korean SME partner currently operates a fleet of internal combustion engine (ICE) vehicles and is seeking advanced analysis tools to assess the transition to electric vehicles (EVs). The partner has an operational full-stack of ICE vehicles and aims to (in the future) convert to a full fleet of EVs. TNO will contribute with algorithms that will enable this assessment. Through joint R&D, these technologies can be quickly integrated into deployable software solutions and validated using real-world data from both Korean and European ICE fleet use cases, supporting effective transition planning and operational feasibility analysis.

Why TIPA: International joint development de-risks market entry and standardisation, and accelerates transfer from research prototypes to a commercial-grade toolchain.</td></tr>
<tr><td>**Key Words**</td><td>Digital twin; charge planning; fleet optimization; smart charging; battery aging</td></tr>
</table>

3. Research and Development Objectives & Plan

<table>
<tr><td>**Technology Readiness Level (TRL)**[2)]</td><td>**Start of the Project**</td><td>Level (4)</td><td>**End of the Project**</td><td>Level (6)</td></tr>
<tr><td>**R&D Objectives**</td><td colspan="4">Objective 1: Assess the energy requirements of the current internal combustion engine (ICE) fleet to establish a baseline for future electrification planning.

Objective 2: Evaluate the possible charging infrastructure and operational capabilities of the current operator to determine readiness for supporting electric vehicles.

Objective 3: Analyze potential charging scenarios and strategies for a fleet that has been converted from ICE to electric, taking into account operational constraints and grid capacity.</td></tr>
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R&D Target Specifications	• Target KPI 1 (prediction): Predict the energy requirements of the future EV fleet within an error margin of 5%, ensuring reliable forecasts for electrification planning [R2]. • Target KPI 2 (operational costs): Provide a comparative analysis of operational costs between the current ICE fleet and the proposed EV fleet, quantifying savings and efficiencies using actual fleet data. • Target KPI 3 (tools): Co-develop at least two tools for energy assessment, leveraging digital twin technology and charge planning solutions to support comprehensive fleet electrification strategies [R1,R9].
R&D Plan	Assumptions (for planning): 36 months total duration, structured in 3 phases (P1: months 1-12, P2: months 13-24, P3: months 25-36). These will be aligned with the final TIPA workplan. The following plan outlines a structured approach for the development and testing of the conversion framework, integrating key phases: requirements gathering, data collection, vehicle selection, energy needs estimation (leveraging digital twin technology), charging infrastructure evaluation, operational assessment, and a concluding phase. The project is scheduled over 36 months, organized into three major phases (P1: Months 1–12, P2: Months 13–24, P3: Months 25–36). Phase 1: Requirements Gathering & Data Collection (Months 1–12) • Requirements Gathering: Engage stakeholders to define operational, technical, and regulatory requirements for fleet conversion. Identify constraints, success metrics, and integration needs with existing systems. • Data Collection: Collect comprehensive datasets on current fleet operations, including vehicle usage patterns, routes, grid asset availability (grid connection, BESS, etc). Phase 2: Vehicle Selection & Energy Needs Estimation (Months 13–24) • Vehicle Selection: Analyze collected data to select candidate vehicles for conversion, prioritizing those best aligned with electrification goals, operational feasibility, and cost-effectiveness. • Energy Needs Estimation (Digital Twins): Develop and calibrate digital twin models for selected vehicles to predict energy consumption profiles under various operational scenarios. Phase 3: Charging Infrastructure assessment, Operational Assessment & Conclusion (Months 25–36) • Charging Infrastructure Dimensioning: Use digital twin outputs and operational forecasts to determine optimal charging infrastructure requirements, considering grid constraints, charger compatibility, and anticipated peak demand. • Renewables and energy storage: asset the impact that renewable energy production

	(e.g., solar panels or wind generation) would have in the fleet conversion. That implies a technical and an economical assessment of the impact of adding these technologies. • Operational Assessment: Pilot the conversion framework in a real-world or simulated environment. Monitor key performance indicators (KPIs) such as cost savings, demand peaks, schedule feasibility, and prediction accuracy. Stress-test system robustness under realistic operational disturbances. • Concluding Phase: Finalize documentation, deliver the software package (including API specifications and reference implementations), and provide deployment support. Summarize lessons learned and recommendations for scaling and future development. This phased approach ensures that all necessary aspects—from initial requirements through to operational validation and delivery—are systematically addressed for a successful and scalable fleet conversion framework.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) responsibilities: • Provide scientific leadership in the fields of charge planning and energy estimation for fleet electrification initiatives. • Offer reference modeling techniques for both charge planning optimization and energy needs estimation, leveraging advanced digital twin approaches. • Develop and validate innovative methods for vehicle digital twin related energy estimation solutions. • Support dissemination efforts through publications, workshops, and contributions to interoperable interface specifications, ensuring knowledge transfer and industry engagement.
Expected R&R of Korean SME Partner	Korean SME partner responsibilities: • Provide fleet planning data for internal combustion engine (ICE) vehicles to support analysis and transition planning. • Support the selection of feasible charging solutions and electric vehicle (EV) infrastructure tailored to operational needs, using fleet planning data provided by the SME on which specific chargers and electric vehicles are available to SME partner. This ensures that the dimensioning and recommendations are based on realistic and accessible assets. • • Analyze the business case and resulting EV fleet operational data to inform strategic decisions. • Lead business case development, customer engagement, and commercialization planning for both the Korean and international markets.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected Technical Outcomes • Development of a comprehensive toolset to support the conversion of internal combustion engine (ICE) fleets to electric vehicle (EV) fleets, incorporating data-driven planning modules and decision-support interfaces. • Delivery of an assessment report providing a detailed techno-economic analysis of the most viable ICE to EV fleet conversion routes, including operational data, cost projections, and feasibility criteria. Expected Impact • Accelerate the transition to commercial operation of EV fleets by providing actionable insights and practical tools for fleet operators. • Strengthen collaboration between Korea and the Netherlands, fostering the development of standardized, interoperable smart charging solutions. • Achieve significant CO2 reductions through the conversion of existing vehicle fleets to electric mobility.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization